

AYUSHI CHADHA

RESEARCH ENGINEER · LATENT & RECURRENT REASONING · META-CONTROL · AGENTIC SYSTEMS

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SUMMARY

AI researcher and engineer working on latent and recurrent reasoning since 2025, following production work on LLM and agentic systems. Research explores adaptive computation and meta-level control, progressing from how internal subgoals persist to whether learned revision policies outperform fixed schedules. Sole-author work accepted at **CompLearn @ ICML 2026**, with follow-up research under review at the NeurIPS 2026 Workshop on Responsible Use of Meta-Agents.

RESEARCH

When to Re-Plan: Subgoal Persistence in Hierarchical Latent Reasoning

ACCEPTED · CompLearn @ ICML 2026 · Sole author

Jan 2025 - May 2026 · arXiv:2606.03741

Studied how long an internal directional subgoal should persist before revision in hierarchical latent reasoning. Extended HRM with a feudal-style manager-worker interface, treating persistence length as an explicit variable governing internal reasoning dynamics.

- **Recurrent latent mechanism.** Implemented L2-normalised directional subgoals, modulated by a learned commitment gate and injected into worker hidden-state updates; cosine alignment trains worker displacement over each commitment window.
- **Core result.** Identified persistence period P as the governing control variable. At alignment weight $\lambda=0.05$, moderate P in $[3,6]$ outperformed single-step re-planning and long horizons; $P=1$ fell below the no-subgoal baseline.
- **Controlled ablations.** Full, no-subgoal, and random-direction experiments isolated interference from learned directional content rather than added capacity or auxiliary loss. Additional sweeps characterised the alignment-weight optimum and the roles of normalisation and commitment gating.

Beyond the Clock: Measuring the Value of Adaptive Revision

UNDER REVIEW · NeurIPS 2026 Meta-Agents Workshop · Sole author

May 2026 - Present · arXiv:2609.00874

Extended fixed persistence to learned state-conditioned meta-control: a supervisory controller uses the reasoner's internal trajectory to preserve or replace latent commitments, testing adaptive revision against strong fixed schedules under matched compute.

- **Adaptive controller evaluation.** Trained revision controllers across three precommitted seeds with fixed $M=8$ compute and calibrated intervention budgets. Policies ranged from near-fixed early timing to high-entropy, state-conditioned schedules; none beat the best forced timing policy on its own frozen checkpoint.
- **Causal diagnosis.** Changing only the second revision position raised seed-0 episode-averaged token accuracy from **46.70% to 48.75%**, isolating a timing failure. Matched PERSIST/REPLAN counterfactuals distinguished predictive scores, adaptive decisions, and actual decision value.
- **Value of adaptive timing.** Enumerated all $K=2$ revision schedules. The best fixed schedule captured **71% of the gain from uniform-random timing to a per-episode oracle**. Remaining adaptive headroom was 0.135 percentage points in micro accuracy, concentrated in 7.6% of episodes.

Peer Review

- **NeurIPS 2026 Workshop on Responsible Use of Meta-Agents — Invited reviewer.** Nominated to review research on supervisory systems that monitor, coordinate, evaluate, or manage other agents.
- **CompLearn @ ICML 2026 — Reviewer.** Reviewed research on compositional learning: reusable structures for generalisation and reasoning.

PROFESSIONAL EXPERIENCE

Propero Consulting · Pune, India

Aug 2022 - Dec 2024

Software Developer · ML & Applied AI Engineering

- **ShopiBot — Architecture & technical leadership.** Spearheaded the architecture and development of a domain-specific AI assistant for Shopify developers building stores for partners and clients, supporting technical Q&A, requirements understanding, and implementation guidance.
- **Retrieval & reasoning pipelines.** Led technical development over several months using LangChain, dense retrieval, RAPTOR, and corrective/adaptive RAG. Reduced latency from **53s to 20s (~62%)**, improved top-k precision by **28%**, and reduced hallucinations by **35%**.
- **Agentic workflows & automation.** Developed Active-RAG workflows integrating knowledge graphs, LangChain planners, multi-tool reasoning, and self-reflection. Subsequently experimented with Sema4.ai's automation framework, connecting ShopiBot with LLM tool calling for domain-aware developer workflows.
- **Multimodal AI.** Built an image-generation system using CLIP-guided prompt augmentation, a DreamBooth fine-tuned Stable Diffusion model, and controllable image-to-image translation.
- **Automated testing & evaluation.** Deployed an anomaly-aware checkout-flow tester using DOM-tree pattern mining and RL-style feedback, catching **92% of regressions pre-release**; incorporated LangSmith telemetry for continuous evaluation.

Unify · London, UK · YC-backed

Jul - Sep 2024

LLM Engineer Contributor

- Selected for Unify's contributor programme, collaborating with the core team on a community-built unified LLM query router across model providers.

INDEPENDENT VENTURE

Saint | Instinct

2026 - Present

Product development in progress

- Developing Instinct, an evaluation tool for agents and reasoning systems, to assess when to persist, retry, or re-plan and whether those decisions improve final outcomes. Grounded in original research on subgoal persistence and supervisory decision value.

SELECTED ENGINEERING PROJECTS

Autograd Engine & GPT From Scratch

2024

- Built a minimal autograd engine, trained bigram and trigram models, and implemented a GPT-2-style language model with a custom tokeniser.

LLMs in C and CUDA

2025

- Implemented matrix multiplication, GELU, and layer normalisation in C and CUDA to study the operations underlying transformer models.

TECHNICAL SKILLS

Reasoning & Planning: HRM, recurrent and hierarchical latent reasoning, meta-level control, adaptive computation, test-time compute, hierarchical and feudal RL, options framework.

Languages & Frameworks: Python, C, C++, PyTorch, HuggingFace Transformers, LangChain, LlamaIndex; foundational CUDA.

Infrastructure & Evaluation: Mixed precision, gradient checkpointing, single-node multi-GPU training (T4×2 / L4), Docker, AWS (S3, Lambda, EKS), Pinecone, Chroma, LangSmith.

EDUCATION

Northeastern University, Boston

Fall 2026

MS, Artificial Intelligence — Machine Learning concentration

Admitted · International Impact Award (**30% tuition**). Did not enrol; prioritised full-time research.

G.B. Pant University, Pantnagar

2017 - 2021

B.Tech, Electrical Engineering · 73.2% (First Division).